

RAZGRAD, Bulgaria

WMO: 155490

Lat: 43.567N

Lon: 26.500E

Elev: 1138

StdP: 14.10

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	14.4	6.4	8.0	11.6	11.0	10.0	16.1	36.7	31.7	35.2	31.4	7.9	50	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.2	90.5	73.0	87.5	71.4	84.7	69.7	76.0	87.0	73.4	83.9	71.3	81.5	6.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	126.0	82.9	69.9	114.9	79.9	67.8	106.6	77.4	40.3	86.8	37.8	84.1	35.9	81.8	92.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.7	20.5	16.6	DB	5.8	95.4	5.7	4.0	1.7	98.3	-1.6	100.7	-4.8	102.9	-8.9	105.8	(4)
(5)			WB	5.3	77.9	5.6	4.9	1.3	81.4	-2.0	84.2	-5.1	87.0	-9.2	90.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	32.0	36.1	43.7	52.7	61.5	68.4	72.7	73.0	64.5	54.2	44.9	35.2		
	DBStd	16.43	10.41	10.78	9.20	7.47	6.36	5.92	5.34	5.61	7.12	8.70	9.57	9.82		
	HDD50	1989	560	398	238	53	4	0	0	0	1	59	209	467		
	HDD65	5023	1022	808	661	374	144	33	6	7	94	346	603	925		
	CDD50	3205	3	9	42	134	360	552	704	712	436	188	57	8		
	CDD65	764	0	0	0	5	35	135	245	254	79	10	1	0		
	CDH74	6221	0	0	2	38	256	1041	2095	2165	560	62	2	0		
	CDH80	2016	0	0	0	2	27	285	760	789	145	8	0	0		
	WSAvg	6.6	6.8	7.5	8.1	7.3	6.3	5.6	5.9	5.7	6.3	6.0	6.5	6.7		
	PrecAvg	25.6	1.7	1.4	1.8	2.0	2.6	2.8	2.6	2.0	2.8	2.1	2.0	2.0		
(15) Precipitation	PrecMax	41.9	3.7	4.0	3.6	5.1	6.9	6.8	7.8	6.5	8.6	4.1	4.2	5.5		
	PrecMin	14.9	0.3	0.1	0.1	0.2	0.4	0.4	0.3	0.0	0.0	0.0	0.0	0.1		
	PrecStd	6.3	0.9	0.8	0.9	0.9	1.6	1.4	2.1	1.5	2.1	1.1	1.1	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.4	65.0	71.2	78.7	83.2	90.1	96.6	94.7	88.5	80.3	71.7	60.6		
		MCWB	49.8	52.7	56.0	63.3	66.5	73.4	77.9	73.8	70.5	64.4	59.9	51.5		
	2%	DB	53.4	59.1	66.3	73.2	79.6	86.4	90.6	90.8	84.1	74.7	65.9	56.1		
		MCWB	47.0	49.6	53.6	59.9	65.0	72.1	73.4	72.9	67.1	61.8	57.3	49.9		
	5%	DB	49.9	54.7	62.1	68.5	76.8	83.0	87.4	87.4	80.3	70.5	62.0	52.1		
		MCWB	45.0	47.4	51.0	56.9	63.5	69.5	71.5	71.1	65.5	59.9	54.7	46.9		
	10%	DB	46.2	50.6	57.7	64.8	73.7	80.1	84.0	84.5	76.5	66.4	57.9	48.6		
		MCWB	42.3	44.8	48.6	54.9	62.0	67.8	70.1	69.4	63.7	57.7	52.5	44.5		
	0.4%	WB	50.9	53.9	57.9	64.8	69.3	78.5	83.3	78.1	72.2	65.7	61.8	54.0		
		MCDB	55.9	60.9	68.2	76.5	79.0	87.2	92.6	89.9	86.2	77.6	67.5	58.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.7	50.3	54.4	60.9	66.8	73.8	76.6	75.2	69.1	62.9	58.1	50.6		
		MCDB	52.6	57.9	64.0	71.3	76.7	83.4	86.7	87.8	80.7	72.7	64.8	55.0		
	5%	WB	45.3	47.7	51.8	58.0	64.8	71.0	73.6	73.0	66.8	60.7	55.4	47.4		
		MCDB	49.5	54.0	60.8	66.8	74.4	80.4	83.4	83.8	76.4	68.7	60.8	51.5		
	10%	WB	42.4	45.0	49.3	55.6	62.8	68.9	71.2	70.9	64.8	58.6	53.0	44.7		
		MCDB	45.9	50.1	57.1	63.1	71.8	77.8	80.9	81.4	74.0	65.0	57.3	48.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	12.6	14.6	16.2	17.5	17.4	18.2	18.2	16.7	14.3	11.8	10.4		
		MCDBR	15.7	18.8	20.7	21.5	21.4	21.0	22.6	21.8	22.0	20.0	17.0	15.0		
		MCWBR	10.2	11.5	11.8	11.9	11.3	10.3	10.2	9.3	9.2	9.3	9.5	10.1		
		MCDBR	15.0	17.9	19.4	19.5	20.2	19.6	20.8	20.3	20.3	18.8	15.7	13.9		
	5% WB	MCWBR	10.2	11.4	11.7	11.5	11.5	10.8	11.9	10.1	9.8	9.7	9.2	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.361	0.397	0.440	0.416	0.423	0.431	0.443	0.413	0.381	0.354	0.333		
	taud		2.316	2.251	2.203	2.091	2.232	2.244	2.228	2.194	2.248	2.323	2.351	2.330		
	Ebn at Noon		248	259	265	262	272	270	266	258	256	249	237	234		
	Edh at Noon		27	34	40	49	43	43	43	43	38	31	26	24		
(44) All-Sky Solar Radiation	RadAvg		473	713	1068	1475	1837	2013	2054	1845	1342	869	513	394		
	RadStd		72	99	99	166	145	153	145	100	109	90	69	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.90	N/A	N/A	N/A	+2.47	+3.08	N/A	-418	N/A	N/A
(47)	Regional (45 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	-196	+140	+88

Nomenclature: See separate page